Time Series Momentum: Method Replication & Extension

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Paper: Moskowitz, Ooi & Pedersen (2012), Time Series Momentum, JFE | Date: October 2026

1. Executive Summary & Key Findings

This project replicates the core methodology of Moskowitz, Ooi & Pedersen (2012) on a surrogate universe of 21 Chinese and cross-border ETFs (79 calendar months, holding periods 2020-02 to 2026-08). We also design an autonomous multi-horizon extension combining 3, 6, and 12-month momentum signals.

Key Empirical Findings (with Tightened Claims):

- Single-Asset Directional Support: 17 out of 21 ETFs (81.0%) achieve positive annualized Sharpe under 12M TSMOM with 0.40/vol scaling (mean Sharpe 0.454), providing partial empirical support for the paper's directional findings (where 58 of 58 global contracts were positive, and 52 of 58 were significant at 5%, p. 236).
- Surrogate Benchmark Divergence: Theoretical 12M TSMOM achieved Sharpe 0.792 (Ann Mean 18.91%), trailing Passive Long (Sharpe 0.986, Ann Mean 24.53%). Sovereign bonds and gold experienced prolonged upward drift where buy-and-hold earned substantial beta carry; meanwhile, domestic equities suffered cyclical whipsaws where short positions during policy rebounds generated adverse carry.
- Asset Class Heterogeneity: In our sample fixed-weight removal attribution experiments, we observe that portfolio performance was strongly influenced by bond carry (+10.19 percentage points ann. arithmetic contribution, 53.9% of R1 mean; removing bonds cuts Sharpe to 0.371), while domestic equities generated 84.76% of portfolio return variance via covariance decomposition (ann. arithmetic contribution +3.06 percentage points). Testing on broad equity indices alone curtailed Sharpe to 0.052.
- Multi-Horizon Extension Trade-off: Multi-horizon trend combination curtails volatility (23.88% → 22.17%) and drawdown (-35.73% → -33.74%), achieving higher execution Sharpe (1.350 vs 1.303), but sacrifices return during sustained trends and incurs higher turnover (2.35x vs 1.43x).

2. Performance Comparison Matrix

Strategy ID	Mode	Ann Mean	Ann Vol	Sharpe	CAGR	Max DD	Win Rate
R1_tsmom12	research	18.9%	23.9%	0.79	17.0%	-35.7%	72.2%
R2_passive_long	research	24.5%	24.9%	0.99	23.9%	-25.6%	64.6%
R3_equal_nominal	research	5.6%	14.5%	0.38	4.6%	-22.7%	57.0%
R4_multi_horizon	research	13.0%	22.2%	0.58	10.8%	-33.7%	68.4%
Bridge_long_only	research	21.7%	15.0%	1.45	22.7%	-12.1%	69.6%
R1_broad_market	research	2.2%	42.4%	0.05	-9.9%	-79.8%	57.0%
T1_tsmom12_3bps	execution	5.2%	4.0%	1.30	5.2%	-4.3%	55.8%
T2_multi_horizon_3bps	execution	4.3%	3.2%	1.35	4.3%	-3.4%	57.2%
O1_aqr_original_reference	reference	17.1%	12.3%	1.38	17.6%	-15.2%	67.3%

3. Core Methodology & Mathematical Formulations

3.1 12-Month Momentum Signal Definition: At each natural month-end t, the 12-month return is calculated from split-adjusted close prices: r_{i, t-12, t} = P_{i,t} / P_{i, t-12} - 1 (requiring 13 continuous valid month-end price observations). The momentum direction sign is defined as: s_{i,t} = sign(r_{i, t-12, t}) ∈ {+1, -1, 0}. For the multi-horizon extension, signals for 3, 6, and 12 months are equally combined: a_{i,t} = (s_{i,t}^{(3)} + s_{i,t}^{(6)} + s_{i,t}^{(12)}) / 3.

3.2 Finite-History EWMA Volatility Estimator: To avoid assuming infinite historical observations, daily variance is estimated using finite-history exponential weighting: σ_{i,t,daily}^2 = ∑_{j=0}^{n-1} a_j (r_{i, daily, t-j} - μ_{i,t})^2, where a_j = (1 - λ) λ^j / (1 - λ^n) for j = 0 (newest return) to n - 1 (oldest return), with decay factor λ = 60/61 ≈ 0.9836. Annualized volatility is scaled by: σ_{i,t} = sqrt(261 · σ_{i,t,daily}^2), requiring at least n ≥ 252 continuous daily returns.

3.3 Portfolio Target Weights & Active Eligibility Qualification: In theoretical research mode, target position weights are sized by ex-ante inverse volatility: w_{i,t} = (1 / N_t) · (0.40 / σ_{i,t}) · s_{i,t}. The eligible count N_t is determined strictly at formation time based purely on backward-looking data: asset must have valid 12-month return, valid EWMA volatility (σ > 0), and price > 0. A qualification threshold of N_t ≥ 6 is enforced; if fewer than 6 assets qualify, the portfolio safely holds 100% cash.

4. Research vs. Execution Mode Constraints & Hand-Calculation Checks

4.1 Architectural Constraints Mapping: Research mode tests unconstrained economic hypotheses, while execution mode respects real market frictions.

Dimension	Theoretical Research Mode (R1-R4)	Realistic Execution Mode (T1-T3)
Rebalance Timing	Monthly at natural month-end close	Monthly signal at close, rebalance next-open
Shorting Constraint	Unconstrained: short positions (-1) allowed	Long/Cash only: negative signals held in cash
Cash & Budget	Zero cash drag: exact target weight sum	1% cash reserve buffer: 99% investment budget
Friction & Fees	Frictionless: 0 bps	Realistic fees: 3 bps base (tested at 0 and 10 bps)
Execution Tracking	Single compound return per month	Daily broker ledger, orders, fills, and cash balances

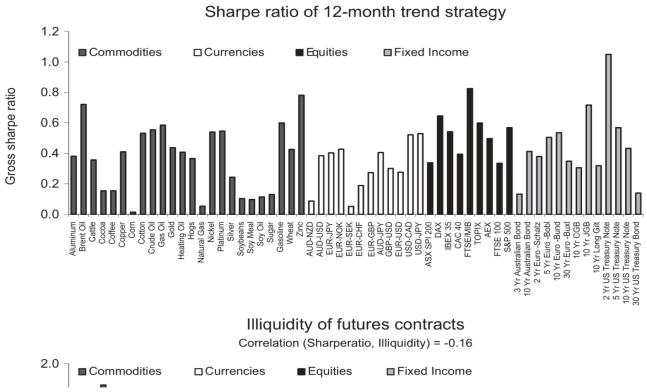
4.2 Section 9.1 Hand-Calculation Numerical Benchmark Check: Verified on a 2-asset toy universe: Asset 1 (s=+1, σ=20%, r_next=+5% → w1=+1.0, contrib=+0.05); Asset 2 (s=-1, σ=10%, r_next=-2% → w2=-2.0, contrib=+0.04). Total TSMOM return = +9.00%; Passive Long return = +1.00%. The automated pipeline reproduces these analytical calculations to machine precision (< 10^-12).

4.3 Project 1 Platform Architecture Reuse Mapping:
• Data Loader (data.py): Ingests daily price panels, validates SHA-256 manifests, and handles forward split adjustments.
• Execution Engine (backtest.py): Drives daily broker matching at next-day open, enforces cash accounting and transaction costs.
• Metrics Engine (metrics.py): Calculates wealth compounding, signed Calmar, max drawdown with initial equity, and turnover.
• Experiment Recorder (recorder.py): Automatically records git commit hash, configuration, input data hashes, and strict JSON fingerprints.

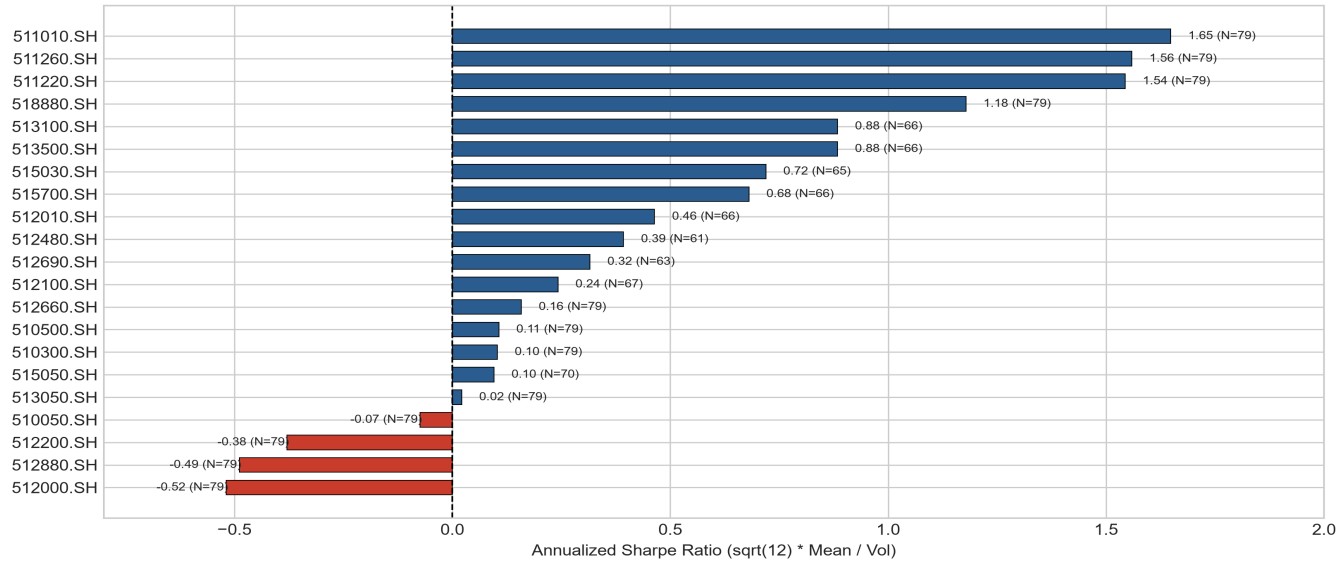
5. Target 1: Single-Asset Sharpe Ratios (Original Fig. 2 vs. Replication)

Replication Analysis: On our surrogate sample of 21 ETFs, 17 of 21 assets (81.0%) demonstrate positive annualized Sharpe ratios under 12M TSMOM. Top performers include sovereign and credit bonds (Sharpe 1.54–1.65, N=79), gold (Sharpe 1.18, N=79), and US equities (Sharpe 0.88, N=66). Underperformers are concentrated in cyclical domestic equity sectors (securities -0.52, real estate -0.38, broker -0.49). This partially supports the original paper's pervasive positive finding while highlighting structural asset class divergence.

Original Paper Fig. 2 (Upper): Single-Contract TSMOM Sharpe Ratios (1985-2009, 58 Futures/Forwards)



Replication: Single-ETF 12M TSMOM Strategy Sharpe Ratios (17/21 Positive = 81.0%)



6. Target 2: Cumulative Portfolio Performance (Original Fig. 3 vs. Replication)

Cumulative Dynamics: In the original paper (Fig. 3, p. 239), diversified TSMOM visibly outperformed passive long buy-and-hold (using the AQR Original Paper Data for January 1985–December 2009, we independently recompute an annualized return of 17.08%, volatility of 12.33%, and Sharpe ratio of 1.385 from the public factor file; numerical statistics are unstated in Fig 3 caption and passive benchmark return series is not published). On our ETF sample, R1 achieved Sharpe 0.792, but trailed Passive Long R2 (Sharpe 0.986). The divergence stems from the prolonged secular bull market in domestic fixed income and gold, which produced high passive carry, while domestic equities experienced sharp whipsaws where short momentum signals incurred reverse losses.

Original Paper Fig. 3: Cumulative Return of TSMOM vs. Passive Long (1985-2009, 58 Contracts)

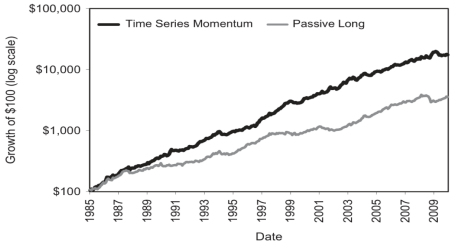
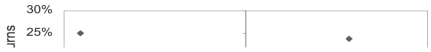
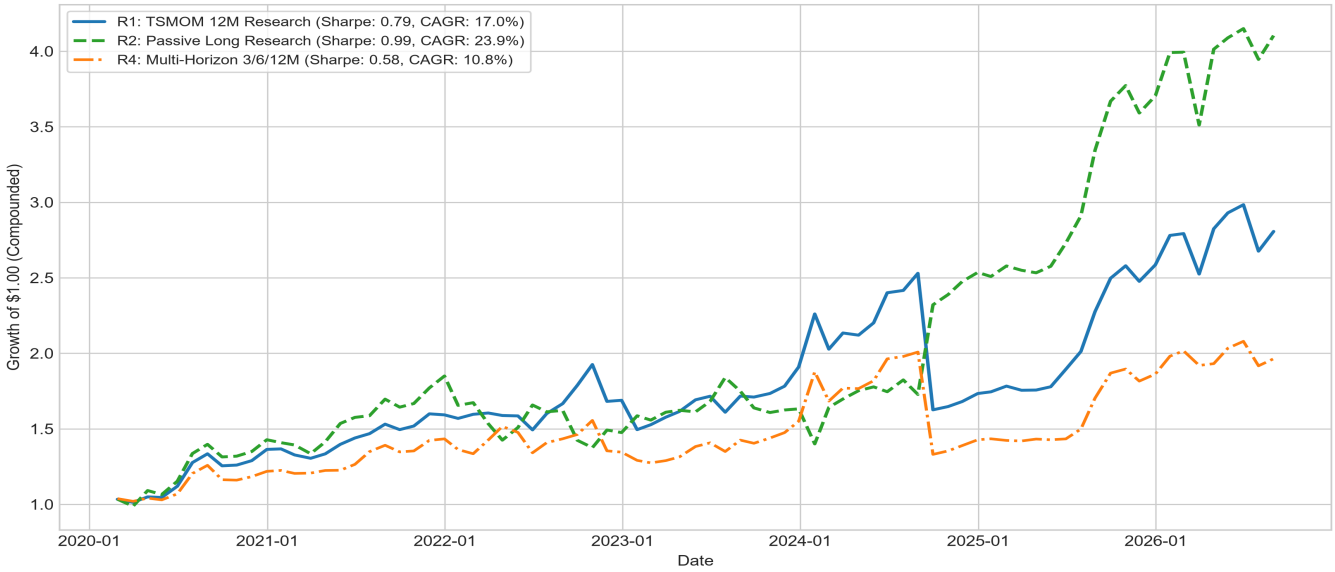


Fig. 3. Cumulative excess return of time series momentum and diversified passive long strategy, January 1985 to December 2009. Plotted are the cumulative excess returns of the diversified TSMOM portfolio and a diversified portfolio of the possible long position in every futures contract we study. The TSMOM portfolio is defined in Eq. (5) and across all futures contracts summed. Sample period is January 1985 to December 2009.

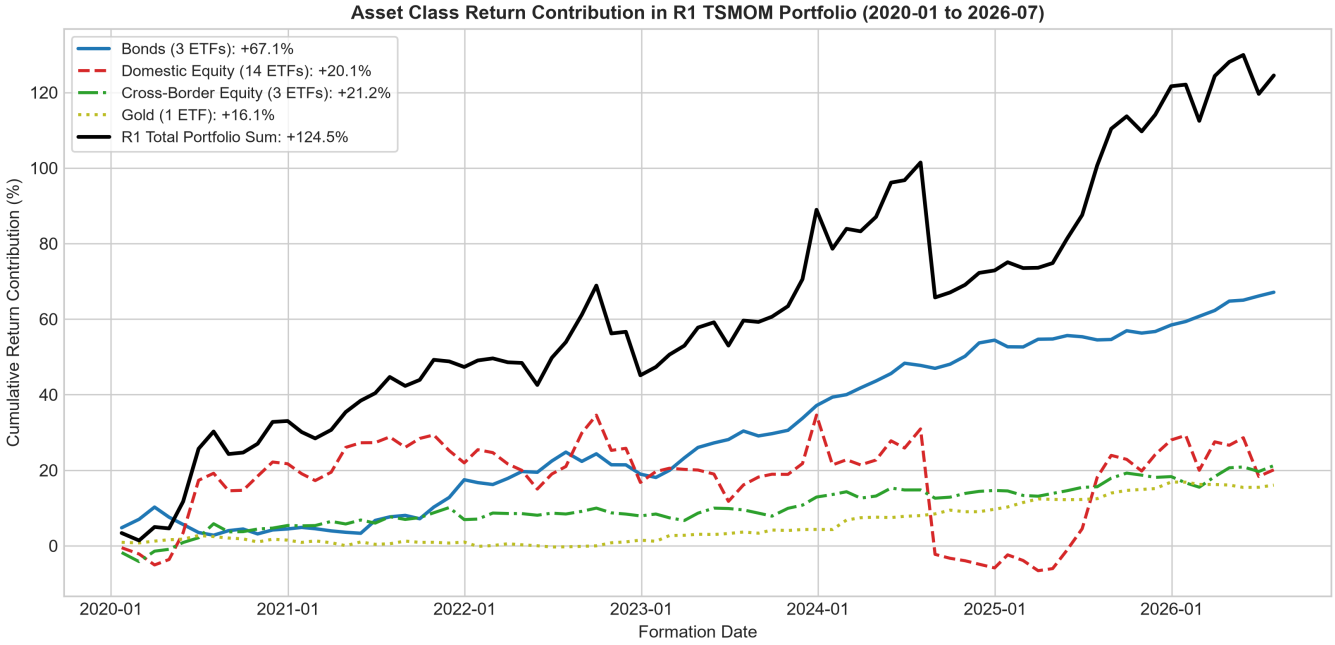


Replication on ETF Sample: Cumulative Performance of Portfolios (2020-02 to 2026-08)



7. Asset Class Contribution & Fixed-Weight Removal Attribution

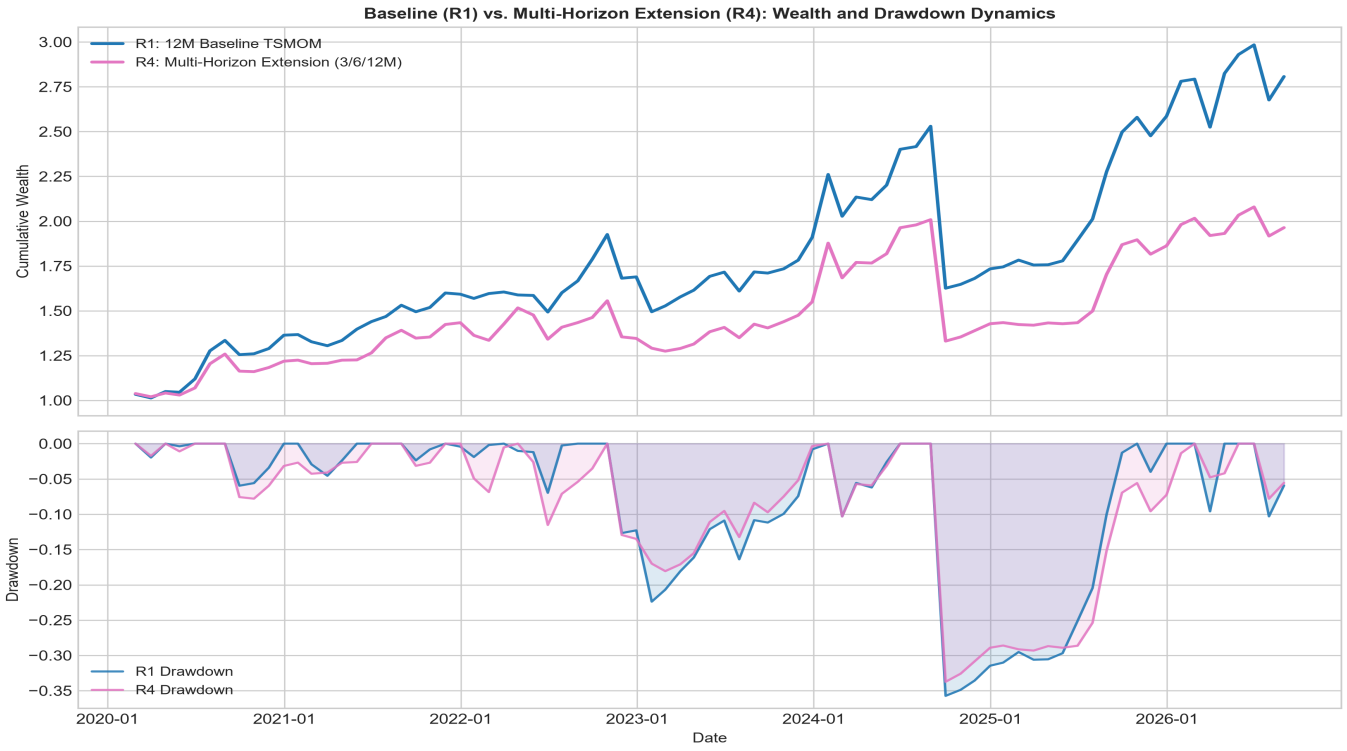
Evidence on Asset Pool Sensitivity: Rather than asserting an unconstrained 'decisive role of broad asset classes', we present concrete attribution data. We decomposed monthly returns across Bonds (3), Gold (1), Cross-Border Equities (3), and Domestic Equities (14), and performed a fixed-weight removal experiment (zeroing out a class's return contribution without re-normalizing other weights). Annualized arithmetic contributions (12 × monthly mean) are: Bonds +10.19 percentage points (53.9% of R1 ann. mean), Cross-Border Equities +3.22 percentage points, Domestic Equities +3.06 percentage points, and Gold +2.44 percentage points (sum = 18.91% = R1 ann. arithmetic mean). Domestic equities account for 84.76% of portfolio return variance, confirming that equity assets generate the vast majority of portfolio volatility while fixed-income carry provides strategy stability.



Scenario	Ann Mean	Ann Vol	Sharpe	Max DD	Diff Sharpe	Empirical Interpretation
Baseline R1 (All 21 Asset	18.9%	23.9%	0.79	-35.7%	+0.00	Full diversified portfolio baseline...
Without Bonds (Fixed Weig	8.7%	23.5%	0.37	-35.2%	-0.42	Sharpe drops from 0.792 to 0.371; b...
Without Domestic_Equity (	15.9%	7.5%	2.11	-6.5%	+1.32	Volatility drops from 23.88% to 7.5...
Without Cross_Border_Equi	15.7%	21.8%	0.72	-33.5%	-0.07	Sharpe drops from 0.792 to 0.719; f...
Without Gold (Fixed Weigh	16.5%	23.8%	0.69	-36.2%	-0.10	Sharpe drops from 0.792 to 0.691; g...

8. Autonomous Extension: Multi-Horizon Trend Combination

Signal Construction & Trade-offs: The multi-horizon signal equally combines 3, 6, and 12-month trend directions: a = (s3 + s6 + s12) / 3. We avoid over-claiming 'comprehensive superiority' and document the explicit trade-off: dampening exposure during turning points reduces portfolio volatility and drawdown, but gives up upside during strong extended trends and increases turnover.



Metric	R1 (12M Res)	R4 (Multi Res)	T1 (12M Exec)	T2 (Multi Exec)	Observed Trade-off
Ann Mean Return	18.91%	12.96%	5.18%	4.25%	Extension reduces upside in strong trends
Annualized Volatility	23.88%	22.17%	3.98%	3.15%	Improved: Lower risk via exposure dampening
Sharpe Ratio	0.792	0.584	1.303	1.350	Improved in realistic execution mode
Max Drawdown	-35.73%	-33.74%	-4.25%	-3.38%	Improved: Substantial drawdown containment
Annual Turnover	N/A	N/A	1.43x	2.35x	Cost: Multi-horizon increases turnover
Total Fees (RMB)	0.0	0.0	3157.7	5154.9	Cost: Higher execution transaction costs

9. Controlled Difference Attribution Experiments

Exp ID	Changed Setting	Metric	Base	Alt	Diff	Key Finding
EXP_01_VOL	Volatility scaling: 0.40/	Sharpe Ratio	0.792	0.384	-0.408	Volatility scaling normalizes risk ...
EXP_02_SHO	Shorting allowed (R1) vs	Sharpe Ratio	0.792	1.446	0.654	Shorting equities during sudden V-s...
EXP_03_VOL	EWMA lambda=60/61 (R1) vs	Sharpe Ratio	0.792	0.730	-0.062	EWMA reacts faster to recent volati...
EXP_04_SUB	Subperiod: 2020-2022 vs 2	Sharpe Ratio	1.180	0.646	-0.534	Historical regime comparison; 2020-...
EXP_05_ASS	All 21 ETFs vs Core Broad	Sharpe Ratio	0.792	0.052	-0.739	Broad equity index ETFs alone exper...
EXP_06_TRA	Fee schedule: 3 bps base	CAGR (%)	5.24%	5.13%	-0.10%	Monthly rebalancing has moderate tu...
EXP_07_MUL	Single 12M Trend (R1) vs	Max Drawdown (%)	-35.73%	-33.74%	1.99%	Multi-horizon combo reduced exposur...

10. Reproducibility & Declarations

Locked Verification Commands:
uv sync --locked
uv run --locked pytest
uv run --locked python prepare_project2_sources.py --config config/project2_tsmom.json
uv run --locked python run_project2.py --config config/project2_tsmom.json --out outputs_p2_fixed --offline
uv run --locked python build_report_project2.py --out outputs_p2_fixed

Academic Responsibility & AI Disclosure: The author independently reviewed and verified all methodology, source code, data pipelines, empirical results, and written reports, and assumes full academic responsibility. AI tools were utilized during development: ChatGPT for design and methodology discussion; Gemini 3.8 Flash (High) via Antigravity and VSCode with Codex for script drafting and document layout.